MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTSU (INCLUDING VARIATION LSTTSA)

This limited coordination military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24643. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, AWG size 22, class K stranding.
Second - Extruded silicone rubber insulation colored (see Fourth). Minimum thickness 0.017 inch.
Third - Insulation jacket of polyamide, type I or type IV of MIL-M-20693, colored (see Fourth). Wall thickness 0.003-0.003-inch, nominal. Diameter over insulated and jacketed conductor shall be 0.076-inch, minimum twisted pair identification code, applied by method 6.
Fourth - Two conductors, one white, one black, twisted together to form a pair having a lay length not greater than 2-1/4 inches. For LSTTSU-1-1/2 only, three conductors, one white, one black, one red, twisted together to form a triad having a lay not greater than 2-1/4 inches.
Fifth - The specified number of pairs (see table I) cabled together with a lay not greater than 10 inches. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers shall be used to form a firm, well-rounded assembly.
Sixth - Binder or combination binder and barrier.

© denotes changes.

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Construction variant for type LSTTSU

Seventh - Cross-linked polyolefin jacket. (See table I for thickness.) Cable surface marking required.

Construction variant for type LSTTSA

Seventh - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking not required.
Eighth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/23	Size 2/	Type's overall diameter (maximum) (inches)		Number of pairs	Cable jacket thickness (nominal) (inch)	Cold bending mandrel diameter (maximum) (inches)
		LSTTSU	LSTTSA			
-01UN	1-1/2	3/ 0.330	0.380	1-1/2	0.050	5
-02UN	3	.450	.500	3	.050	7
-03UN	5	.540	.590	5	.050	8
-04UN	10	.675	.725	10	.062	9
-05UN	15	.800	.850	15	.062	10
-06UN	20	.870	.920	20	.062	11
-07UN	30	1.080	1.130	30	.075	13
-08UN	40	1.200	1.250	40	.075	15
-09UN	50	1.400	1.450	50	.075	16
-10UN	60	1.450	1.500	60	.075	17

1/ Unarmored. When armored option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSTTSU and LSTTSA.
3/ LSTTSU-1-1/2 comprised of three conductors cabled together to form a triad.

IDENTIFICATION CODES:

(a) Twisted pair identification code. This code consists of numbers in sequence running from 1 inch through the number corresponding to the total quantity of twisted pairs in the cable. Both conductors in each pair shall be numbered the same, denoting the sequence number of the pair. Distinction between the two conductors is provided by different colored insulation.

(b) Twisted triad identification. This code consists of numbers in sequence running from 1 inch through the number corresponding to the total quantity of twisted triads in the cable. All three conductors shall be numbered the same, denoting the sequence number of the triad; distinction between the three conductors is provided by different colored insulation.

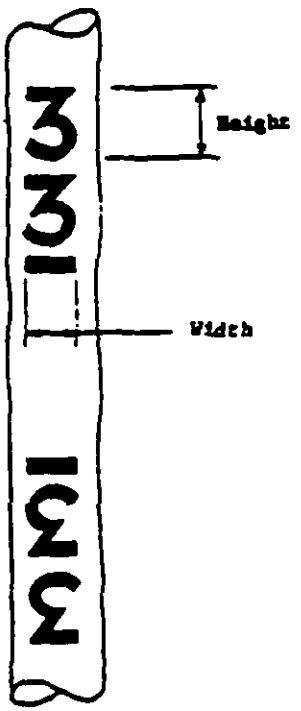
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IDENTIFICATION METHOD:

(a) Method 6. Identification method 6 shall consist of numerals printed in ink on the conductor insulation. For conductors having a jacket directly over the insulation, the numerals may be printed in ink on the jacket, at the manufacturer's option. White ink shall be used for a red or black background; black ink shall be used for a white background. Numerals shall be upright on the longitudinal axis of the conductor (see figure 1). Numeral width shall be proportional to conductor, o.d. as follows:

Conductor (o.d.) (inch)	Numeral width (inch)
0.040 - 0.050	0.020
.051 - .070	.025
.071 - .095	.030

Numeral height shall be 2-1/2 to 3 times numeral width. Each numeric legend shall be underlined. Two-digit legends shall have the bottom numeral underlined. Legends shall be alternately inverted and shall be repeated at intervals not greater than 1-1/2 inch.



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FIGURE 1. Conductor of pair number 33.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	18.60
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to ground	1000
Insulation resistance - megohms/1000 feet, minimum	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Crack resistance	No damage
Group B:	
Gas flame - 1 hour (see note 1)	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure
Armor (type LSTTSA only) - conformance to material, construction and coverage	No failure

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NOTE: 1. Braided armor, when present, shall be connected to neutral terminal of power supply. For gas flame test of LSTTSU-1-1/2, the three conductors shall be connected to phases A, B and C of the power supply. For test of all other sizes, connection of conductor pairs shall be in accordance with the following pattern:

(a) Pair #1: One conductor to phase A, one conductor to phase B.
(b) Pair #2: One conductor to phase C, one conductor to phase A.
(c) Pair #3: One conductor to phase B, one conductor to phase C.
(d) Pair #4: One conductor to phase A, one conductor to phase B.
(e) Pair #5: One conductor to phase C, one conductor to phase A.
(f) Pair #6 through Pair #60: In similar sequence.

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Gas flame 3 hour (see note 1)	No failure
Cold working (minus 20 ± 2°C)	No damage
Specific gravity of extruded silicone insulation maximum	1.55
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTTSU-1-1/2 through -10	2500
LSTTSU-15 and -20	1500
LSTTSU-30 through -60	1000

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-N331)

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